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Subject – C Programming

100 DAYS OF CODE - 4

(C PROGRAMMING)

❖ Problem Statement:

Correct the given C program, algorithm, and flowchart to perform multiplication of two integers without using the multiplication (*) operator. Implement the solution using repeated addition and ensure that it correctly handles positive numbers, negative numbers, and zero.

• Objective:

To implement multiplication of two integers using repeated addition instead of the multiplication (*) operator and verify the program using different test cases.

• Given Algorithm:

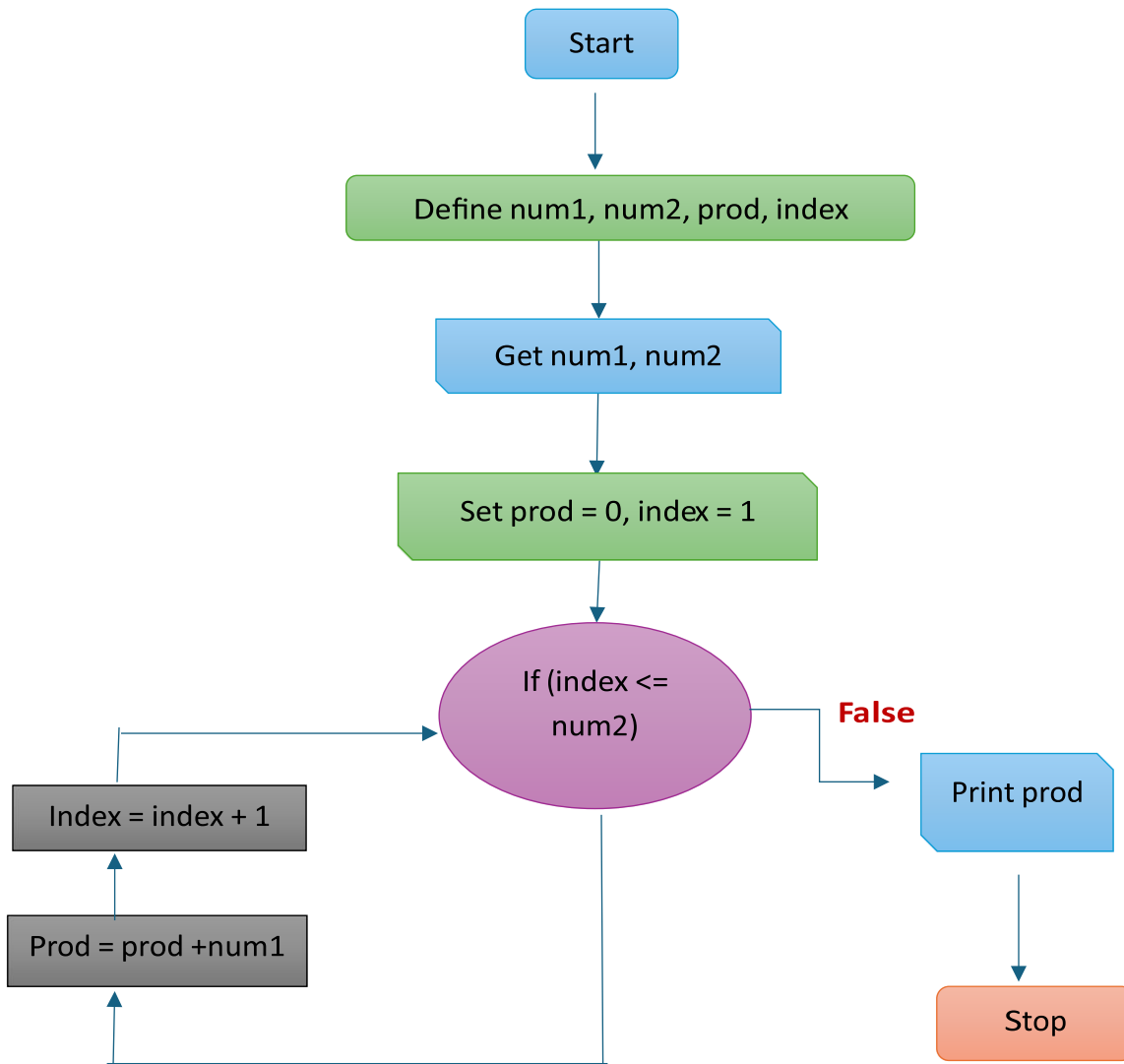
- 1) Define num1, num2, prod, index as integers.
- 2) Get num1, num2.
- 3) Set prod = 0, index = 1
- 4) Repeat steps 4.1 and 4.2 until (index <= num2)
 - 4.1 prod = prod + num1
 - 4.2 index = index + 1
- 5) Print prod
- 6) Stop

○ Here are the mistakes in the given algorithm:

- 1) It works only for positive values of num2.
- 2) It does not work for negative numbers.
- 3) It does not correctly handle multiplication if one of the inputs is zero.

4) It does not determine the sign of the result.

- **Flowchart**

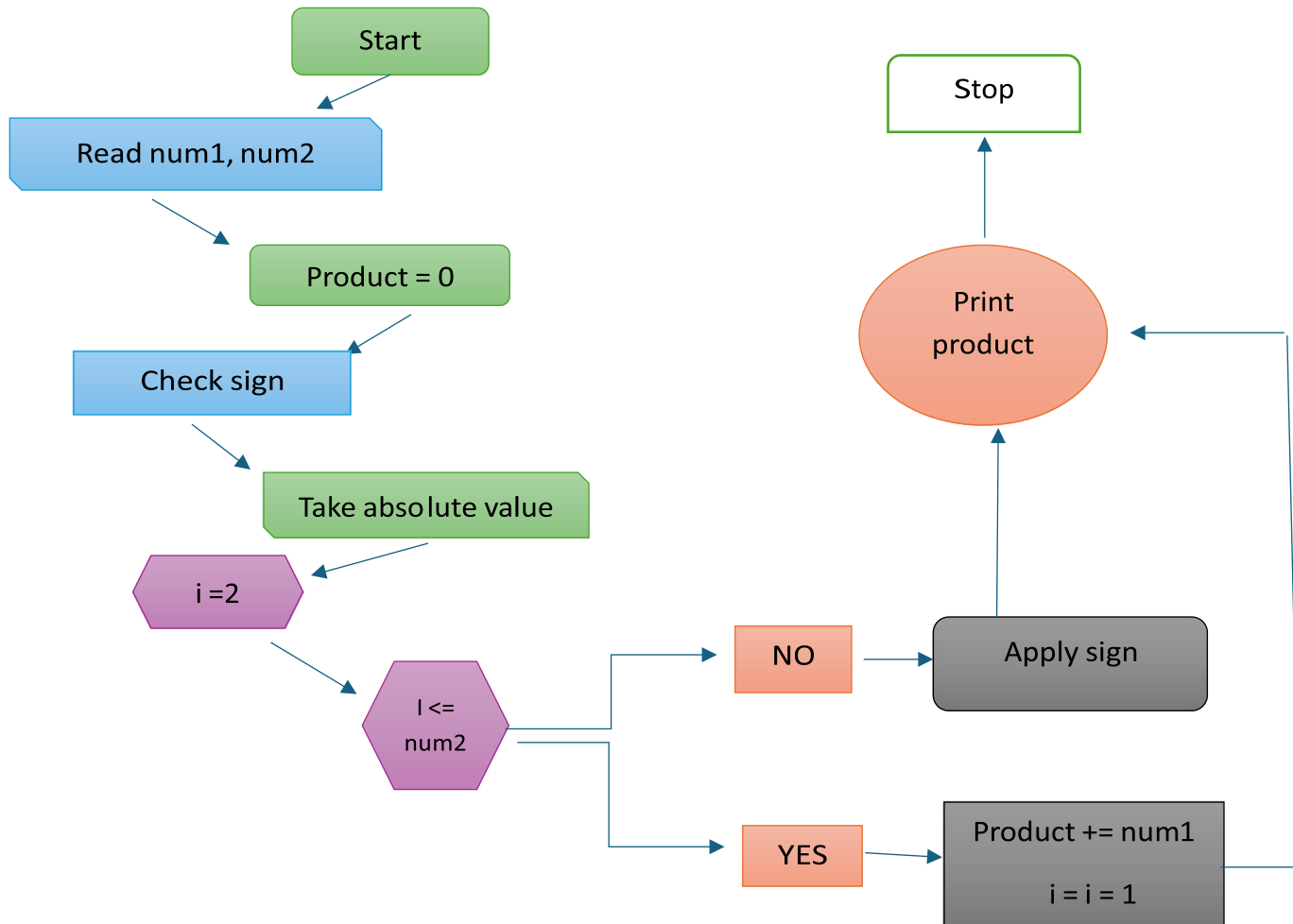


○ Here is the correct algorithm:

1. Start.
2. Read two integers num1 and num2.
3. Initialize product = 0.
4. Determine whether the result should be negative.
5. Convert num1 and num2 to their absolute values.
6. Repeat num2 times:

- $\text{product} = \text{product} + \text{num1}$
- 7. If the result should be negative, make $\text{product} = -\text{product}$.
- 8. Display the product.
- 9. Stop.

○ Here is the correct flowchart:



❖ CONCLUSION:

This assignment demonstrated how we can multiply using repeated addition instead of using the $*$ operator. The program was tested with positive numbers, negative numbers and zero and it produced correct outputs every time.